



S.I.G.N.A.L.

Shell Intelligence Generating News Analysis & Learning



USER OPERATION MANUAL

Version 2.0 | Shell Intelligence System

AI-Powered Crude Oil Market Intelligence Platform

Prepared for:	Senior Design Report
Platform:	S.I.G.N.A.L. v2.0
Audience:	Executives, Analysts, Viewers

1. Introduction

S.I.G.N.A.L. (Shell Intelligence **G**enerating **N**ews **A**nalysis & **L**earning) is an AI-powered crude oil market intelligence platform designed to help energy professionals monitor global oil markets, interpret geopolitical news, and make data-backed trading decisions.

The system combines real-time NLP sentiment analysis, machine-learning price forecasting, and a role-gated analytics dashboard into a single unified interface. Two distinct AI models drive the core intelligence:

- **Direction Model** — An ensemble classifier (Random Forest, 81.5% accuracy) that predicts whether WTI crude oil will move UP or DOWN on the next session.
- **Price Forecast Model** — A stacking ensemble (Ridge + HistGBM + LightGBM → RidgeCV meta-learner) that predicts the next closing price in \$/bbl, beating a naïve random-walk baseline (RMSE 2.07 vs 2.09, R² 0.953).

Everything is surfaced through a theme-aware, role-gated dashboard so analysts, executives, and viewers each see exactly the information relevant to their role.

2. System Requirements

Requirement	Version / Notes
Python	3.10 or higher
Node.js	18 or higher
Web Browser	Chrome, Firefox, or Edge (latest)
macOS only	Homebrew with libomp for XGBoost support

3. Starting the Application

S.I.G.N.A.L. has two components that must both be running simultaneously: the **backend API server** and the **frontend dashboard**. Open two separate terminal windows.

3.1 Start the Backend (Terminal 1)

Run all commands from the project root (SrDesign-Shell/):

```
python3 -m venv venv
source venv/bin/activate # Windows: venv\Scripts\activate
pip install -r requirements.txt
uvicorn backend.main:app --reload
```

The backend starts on **http://localhost:8000** and auto-seeds the database on first run. If an external news API is unavailable it falls back to synthetic data automatically.

Note: You must create and activate the virtual environment before running pip or uvicorn. Inside the activated venv the short names python and pip work automatically.

3.2 Start the Frontend (Terminal 2)

```
cd frontend
```

```
npm install
```

```
npm run dev
```

The frontend starts on **http://localhost:3000**. Open this URL in your browser to access the dashboard.

Note: Both services must be running simultaneously. If the backend is unreachable, a red 'Backend Offline' banner appears with startup instructions.

4. User Accounts and Roles

S.I.G.N.A.L. uses a three-tier role-based access control (RBAC) system. Each role determines which pages and features are visible.

Role	Access Level	Description
Executive	Full access	All pages including Signal Engine; executive briefings and escalation protocols. Identified b
Analyst	Partial access	All pages except Signal Engine. Intended for data analysts needing model insights and con
Viewer	Read-only	Command Center, Intelligence Feed, Commodity View, and Settings only.

4.1 Registering an Account

Navigate to the login/registration page and create an account by entering your name, e-mail address, password, and selecting your role. Roles are assigned at registration and determine which sections of the platform are visible in the sidebar.

5. Onboarding

First-time users are guided through a personalized onboarding flow after registration.

5.1 Analyst Onboarding (5 Steps)

1. Welcome greeting — Introduces the platform and the user's role.
2. Commodity focus — Select a default commodity (WTI or Brent crude).
3. Alert configuration — Set price-move and sentiment-spike thresholds.
4. Notification preferences — Choose which alert types to receive.
5. Completion — Confirm settings and enter the dashboard.

5.2 Executive Onboarding (6 Steps)

1. Cinematic security verification — Animated identity and clearance confirmation.
2. Strategic briefing — Live KPI preview (sentiment, model prediction, headline activity).
3. Strategic focus areas — Select oil market regions and event types of interest.
4. Escalation protocols — Configure briefing cadence and escalation level.
5. Dashboard preview — Visual tour of the platform layout.
6. Activation sequence — Confirm credentials and enter the dashboard.

6. Dashboard Layout

The main dashboard consists of three persistent elements:

Element	Location	Description
Sidebar	Left edge	Collapsible navigation menu with links to all accessible pages and a system status indicator.
Top Bar	Top of page	Displays a live UTC clock, Notification Center bell icon, and Command Palette trigger.
Main Content Area	Center/right	Page-specific content rendered here.

6.1 Sidebar

- Click the **chevron arrow** (▢ / ▢) on the right edge to collapse or expand. Collapsed mode shows icon-only buttons.
- System status indicators at the bottom show API Online, Model Active, and News Feed Live — amber (checking) → green (healthy) → red (offline).

6.2 Top Bar

- **Live Clock** — Current UTC time, updated every second.
- **Notification Bell** — Opens the Notification Center dropdown with severity-coded alerts. Mark individual alerts as read or clear all at once.
- **Theme Toggle** — Cycles between Dark, Black, and Light themes.

6.3 Global Command Palette

Press **⌘K** (macOS) or **Ctrl+K** (Windows/Linux) from anywhere in the dashboard to open the Command Palette. Type to search headlines, navigate pages, or run actions (Refresh Data, Logout). Use arrow keys to move between results and Enter to select.

7. Page Reference

7.1 Command Center (/dashboard)

Access: All roles

The Command Center is the main overview page and default landing screen after login.

Components:

- **Greeting Banner** — Personalized greeting with system status, UTC timestamp, WTI / BRENT selector, and Refresh button.
- **KPI Cards (4)** — Avg. 24h Sentiment, Model Confidence, Headlines (24h), High-Impact Alerts.
- **Holdout Chart Panel** — Actual vs. Predicted price chart from the model's 20% out-of-sample holdout test set.
- **Intelligence Feed Panel** — Condensed list of 30 most recent headlines with sentiment badges and impact scores.
- **Price Chart** — Rolling 14-day oil price line chart with news event markers.
- **Alerts Panel** — High-impact headlines that triggered configured alert thresholds.

7.2 Intelligence Feed (/dashboard/feed)

Access: All roles

Full-screen headline browser for all ingested news articles.

Features:

- **Search Bar** — Filter headlines by keyword in title or source.
- **Event Type Filter** — All / Geopolitics / Supply / Demand / Macro / Weather / Regulatory.
- **Headline Cards** — Title, source, publication time, sentiment badge (green / red / amber), AI prediction label (UP ↑ / DOWN ↓ with confidence %), impact score, and external link.
- **Download Button** — Exports the current filtered headline list.
- **Refresh Button** — Re-fetches the latest headlines from the backend.

7.3 Commodity View (/dashboard/commodity)

Access: All roles

Detailed commodity price analysis with multi-panel layout.

Features:

- **Commodity Selector** — Toggle between WTI and BRENT crude oil.
- **Date Range Selector** — View price history over 7 Days, 14 Days, or 30 Days.
- **KPI Cards** — Current price, daily change (\$), daily change (%), and volatility.
- **Price Chart** — Interactive line chart with news event markers. Hover for date, closing price, and related headline.
- **AI Signal Panel** — Current directional prediction (UP/DOWN) with confidence score.
- **Sentiment Breakdown** — Percentage split of positive, negative, and neutral headlines.
- **Recent Headlines** — Latest 10 headlines for the selected commodity.
- **Alerts Panel** — Active high-impact alerts for the selected commodity.
- **Price Statistics** — Period High, Low, Average, and total % Change.

7.4 Data Analytics (/dashboard/analytics)

Access: Analyst, Executive

Statistical analysis and visualization connecting news sentiment to price movements.

Features:

- **Model Performance KPIs** — Accuracy, Precision, Recall, and F1 score from the trained model report.
- **Prediction Distribution Chart** — Bar chart showing historical UP vs. DOWN prediction counts.
- **Event Type Breakdown** — Pie chart of headline proportions by event category, color-coded by type.
- **Sentiment vs. Price Correlation** — Overlaid chart of rolling average sentiment scores against oil price movements.

7.5 Direction Model (/dashboard/model)

Access: Analyst, Executive

Detailed view of the AI direction classifier (UP/DOWN predictor).

Components:

- **Live Prediction Banner** — Current prediction (UP ↑ / DOWN ↓), confidence percentage, and current WTI/Brent price pills.
- **Direction Model Metrics** — Accuracy (81.5%), Precision (83%), Recall (80%), F1 Score (81%).
- **Prediction History Chart** — Time-series of daily predictions showing average confidence, average sentiment, and UP/DOWN counts.
- **Feature Importance Bar Chart** — Relative importance of each input feature, color-coded by category.
- **Collapsible Feature Glossary** — Plain-English explanation of every model input variable. Click to expand.
- **Intelligence Pipeline** — 4-step visual explainer: raw news → sentiment scoring → feature engineering → model prediction.

7.6 Price Forecast (/dashboard/forecast)

Access: Analyst, Executive

Dedicated view of the AI price regression model (next-session closing price predictor).

Components:

- **Confidence Banner** — Color-coded banner indicating model confidence tier (High / Medium / Low) based on recent prediction error.
- **Forecast Metric Cards** — RMSE (\$/bbl vs. baseline), MAE (\$/bbl), R^2 (model fit quality), MAPE — each with plain-English subtitles.
- **Actual vs. Predicted vs. Baseline Chart** — Three-line chart with clickable legend to toggle series on/off. Baseline = naïve last-price (random-walk).
- **Error Analysis Row** — Within $\pm 2\%$, Largest Miss (\$/bbl), Direction Accuracy from the holdout set.
- **Feature Importance Bar Chart** — GBM base-learner importances, color-coded.
- **Collapsible Feature Glossary** — Definitions for every price-model input variable.

- **Model Architecture Explainer** — 3-step description of the stacking ensemble (base learners → meta-learner).
- **Forecast Pipeline** — 4-step pipeline diagram from raw data to final forecast.
- **Model Comparison Table** — Collapsible table comparing all candidate models (RidgeCV, LassoCV, ElasticNetCV, RandomForest, GradientBoosting, XGBoost, Stacking) on RMSE, MAE, R², and MAPE.

7.7 Signal Engine (/dashboard/signals)

Access: Executive only

High-level strategic risk and signal aggregation dashboard for executive decision-makers.

Components:

- **Risk Meter** — Gauge displaying current market risk: STABLE (0–39) / ELEVATED (40–69) / CRITICAL (70–100).
- **Commodity Selector** — Toggle between WTI and BRENT.
- **Market Status Board** — Grid of status indicators (latest price, session prediction, confidence, sentiment trend).
- **Alert Feed** — Chronological list of high-impact alerts with severity coding.
- **Volatility Tracker** — Rolling volatility metric from recent price movements.
- **Refresh Button** — Manually trigger a data refresh.

7.8 Settings (/dashboard/settings)

Access: All roles

User profile management, alert configuration, and application preferences.

Profile Card

Displays user name, e-mail, and role badge (Executive gold / Analyst blue / Viewer grey), role description, and access level.

Alert Configuration

- **Price Move Threshold (%)** — Minimum price percentage change that triggers a price-move alert (e.g., 2.0% means any move $\geq \pm 2\%$ fires an alert).
- **Sentiment Spike Threshold** — Minimum absolute sentiment score for a sentiment alert (0.0 – 1.0).
- **Save Button** — Persists changes. A green checkmark confirms success for 2.5 s.

***Note:** Executive users also have Briefing Cadence (Daily / Weekly) and Escalation Level (Standard / Priority / Critical) fields, set during onboarding.*

Appearance

- **Dark** (default) — Dark charcoal background with Shell yellow/blue accents.
- **Black** — True-black background for maximum contrast or OLED displays.
- **Light** — White/light-grey background for bright offices or presentations.

Theme preference is saved to browser localStorage and persists across sessions.

8. Alerts and Notifications

8.1 Notification Center

Click the **bell icon** in the top bar to open the Notification Center. Alerts are generated automatically when incoming headlines exceed configured thresholds.

Severity	Color	Trigger
Critical	Red	Severe price move or very high-impact geopolitical event.
Warning	Amber	Moderate sentiment spike or elevated impact score.
Info	Blue	New high-activity headline cluster or model prediction change.

Mark individual alerts as read by clicking them, or use **Clear All** to dismiss all at once.

8.2 Configuring Thresholds

Navigate to **Settings** → **Alert Configuration** to adjust when alerts are triggered. Changes take effect immediately after saving.

9. Themes

Theme	Description	Best For
Dark (default)	Dark charcoal backgrounds with Shell yellow/blue accents.	Low-light trading environments
Black	True-black backgrounds, maximum contrast.	Very dark environments or OLED displays
Light	White/light-grey backgrounds.	Bright offices or presentation use

Switch themes using the toggle in the Top Bar or under **Settings** → **Appearance**.

10. Data Export (Power BI Integration)

To export platform data for use in external reporting tools such as Power BI:

Step 1 — Activate the virtual environment:

```
source venv/bin/activate
```

Step 2 — Run the export script from the project root:

```
python3 scripts/export_csv.py
```

Step 3 — Two CSV files are created in the project root:

- headlines_export.csv — All ingested news headlines with sentiment scores, event types, and prediction labels.
- price_points_export.csv — All historical price data points.

Step 4 — Import into Power BI:

In Power BI: **Get Data** → **Text/CSV** → **Upload file** → select each exported CSV → **Next** → **Create**.

11. Troubleshooting

Symptom	Solution
"Cannot reach API" banner	Start the backend: <code>uvicorn backend.main:app --reload</code> from the project root inside an activated virtual environment.
command not found: python or pip	Activate the venv: <code>source venv/bin/activate</code> . Create it first if needed: <code>python3 -m venv venv</code> .
ModuleNotFoundError on backend startup	<code>pip install -r requirements.txt</code> inside the activated venv.
xgboost install fails on macOS	Run <code>brew install libomp</code> before installing requirements.
Dashboard shows no headlines	Click Refresh Data on the Command Center. The system will auto-seed. Alternatively run: <code>curl -X POST http://localhost:8080/api/refresh</code>
Backend crashes with --reload	Run uvicorn from the project root (SrDesign-Shell/), not from inside the backend/ folder.
Onboarding re-appears after registration	Clear browser localStorage (DevTools → Application → Local Storage → Clear All) and register again.

12. System Architecture Overview

Layer	Technology
Frontend	Next.js 16, React 19, Tailwind CSS, Framer Motion, Recharts
Backend API	FastAPI (Python 3.10+), Pydantic, SQLAlchemy, SQLite
AI / Machine Learning	scikit-learn, XGBoost, LightGBM, HistGradientBoosting, VADER Sentiment
Authentication	localStorage session with 3-role RBAC (Executive / Analyst / Viewer)
Theming	3-theme CSS variable system persisted to localStorage